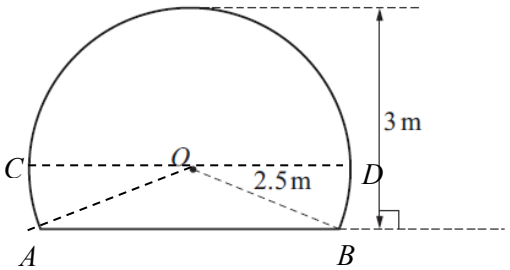
Question	Answer	Marks	Partial Marks	Guidance
23	9835 to 9844 nfww	**7**	**B5** for [area of segment =] 12.29 to 12.31 nfww **M1** for 12.29 to 12.31 × 800 **OR** **B2** for angle at centre = 157 or 156.9 or 156.92 to 156.93 or for [reflex angle =] 203 to 203.1 or **M2** for $2 \times \cos^{-1}\left(\frac{3-2.5}{2.5}\right)$ oe or **M1** for $\cos x = \frac{3-2.5}{2.5}$ oe **M2dep** for $\frac{360-their\theta}{360} \times \pi \times 2.5^2$ or **M1dep** for $\frac{[360-]their\,\theta}{360} \times \pi \times 2.5^2$ seen **M1dep** for $\frac{1}{2} \times 2.5 \times 2.5 \times \sin(their\,\theta)$ oe or $2 \times 0.5 \times \sqrt{2.5^2 - 0.5^2} \times 0.5$ oe **M1** for *their* area × 800 leading to answer	Mark at most accurate In this question for method marks allow 3.14 or $\frac{22}{7}$ for π For B2 accept 2 × 78.5 or 2 × 78.46… or 360 – 2 × 78.5 or 360 – 2 × 78.46… Longer methods possible Implied by major sector angle 203 to 203.1 Implied by x = 78.5 or 78.46… **Dep** 90 < θ < 180 and ≠ 135 M2 implied by 11.1 or 11.06 to 11.08… M2 spoiled if e.g. area of triangle subtracted but allow M1 M1 implied by 8.55 to 8.57 **Dep** 90 < θ < 180 and ≠ 135 and consistent with θ used for area of sector M1 implied by $\frac{\sqrt{6}}{2}$ or 1.22 or 1.221 to 1.225 *their* area must include the area of a sector or area of circle with radius 2.5 cm [± area of a triangle] See AG

Additional guidance Question 23 Alt method apply only if adding sector *AOC* and/or *BOD* to area of semi circle

Question	Answer	Marks	Partial Marks	Guidance
23	9835 to 9844 nfww  	7	**B5** for [area of segment =] 12.29 to 12.31 nfww **M1dep** for (12.29 to 12.31) × 800 **OR** **B2** for angle sectors 2α = 23[.0] to 23.1 or **M2** for $2\times \sin^{-1}\left(\frac{0.5}{2.5}\right)$ soi or **M1** for $\sin\alpha = \frac{0.5}{2.5}$ oe **M2dep** for $2\times\frac{their\alpha}{360}\times\pi\times 2.5^2+\frac{180}{360}\times\pi\times 2.5^2$ or **M1dep** for [2 ×] $\frac{their\alpha}{360}\times\pi\times 2.5^2$ **M1dep** for $\frac{1}{2}\times 2.5\times 2.5\times\sin(180-2\times their\,\alpha)$ oe or $2\times 0.5\times\sqrt{2.5^2-0.5^2}\times 0.5$ oe **M1** for *their* area × 800 leading to the answer	Mark at most accurate In this question for method marks allow 3.14 or $\frac{22}{7}$ for π May be seen later in area calculation α = angle *AOC* or *BOD* M2 may be embedded at the area of sector stage M1 implied by 11.5 to 11.54 M2 and M1dep 0 < α < 45 M2 implied by 11.1 or 11.06 to 11.08… 1.25 or 1.253 to 1.26 nfww or 0.6276 to 0.63[0] nfww M1dep 0 < α < 45 and consistent with α used for area of sector M1 implied by $\frac{\sqrt{6}}{2}$ or 1.22 or 1.221 to 1.225 *their* area must include the area of a sector or area of circle with radius 2.5 cm [± area of a triangle]

General point for both schemes – do not allow e.g. $\frac{3}{5}\pi 2.5^2$ to score M2 or M1 for angle of sector without θ or α stated in method